

Susceptibility and Spatial Signature of Abrupt Thaw Across Alaska's Permafrost Landscapes

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Key points

- Key point 1
- Key point 2
- Key point 3

Abstract

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Plain Language Summary

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1 Introduction

The northern permafrost region contains more organic carbon in its upper three meters of soil than currently resides in Earth’s atmosphere (Hugelius et al., 2014). Most permafrost thaw proceeds by gradual, top-down deepening of the active layer, but abrupt thaw destabilizes and collapses the ground surface over days to years, mobilizing carbon stocks that gradual thaw would take decades to reach (Turetsky et al., 2020; Nitzbon et al., 2020). Under high emissions scenarios, abrupt thaw is projected to impact only 2.5 million km² out of the 18 million km² permafrost region, but threatens to release nearly as much carbon as gradual thaw processes operating over a much larger area (Turetsky et al., 2020). Abrupt thaw leads to a variety of landforms, including thermokarst lakes, collapsing peatlands, and hillslope failures, all of which differ in the quantity and form of carbon released (Olefeldt et al., 2016; Schädel et al., 2016). The scale and timing of carbon release from permafrost soils therefore depend on how and where the landscape undergoes abrupt thaw, and this spatial pattern remains one of the largest sources of uncertainty in the permafrost-carbon feedback (Turetsky et al., 2020; Walter Anthony et al., 2018).

Beyond climate feedbacks, abrupt thaw represents an immediate hazard to the roughly 5 million people who live in the Arctic circumpolar permafrost region (Ramage et al., 2021). Projections suggest that 30-50% of Arctic infrastructure may be at risk of damage by permafrost thaw by 2050, threatening tens of billions of U.S. dollars in damage by the end of the century (Hjort et al., 2022). In Alaska alone, permafrost thaw may cause \$37 to \$51 billion in damage to buildings and roads (Manos et al., 2025). This burden will fall disproportionately on remote and Indigenous communities, whose unofficial, “vernacular” infrastructure and assets often lie outside conventional damage inventories (Verstraete & Zachwieja, 2025). Between 1974 and 2019, thermokarst at Point Lay, Alaska, expanded nearly an order of magnitude faster beneath the developed village than in the surrounding tundra, indicating a potential positive feedback between the built environment and

abrupt thaw (Jones et al., 2025). Predicting how and where abrupt thaw will develop on the landscape is therefore critical for helping Arctic communities adapt to or mitigate the effects of future climate change.

Abrupt permafrost thaw proceeds faster than gradual thaw, by definition, but also results from different processes, leaves distinct surface expressions, and has specific consequences for ecosystems, hydrology, and carbon release (Webb et al., 2025). Gradual thaw typically proceeds from the top down as a progressive deepening of the seasonally thawed active layer into near-surface permafrost (Walter Anthony et al., 2018; Webb et al., 2025). Abrupt thaw, by contrast, is not confined to this vertically ordered progression: subsidence, erosion, mass wasting, and the flow of water and heat can rapidly propagate thaw downward and laterally, affecting deeper or adjacent permafrost (Turetsky et al., 2020; Nitzbon et al., 2020; Webb et al., 2025). The resulting changes to the landscape occur quickly relative to geomorphic time scales, from vertical subsidence of almost a meter over a decade (Farquharson et al., 2019) to thaw-slump headwall retreat of more than ten meters in a single year (S. Kokelj et al., 2015). A site's thaw mode, abrupt or gradual, is not dictated by the presence of permafrost alone, but emerges from an interplay of ground ice, geomorphology, hydrology, and disturbance history (Jorgenson & Osterkamp, 2005; S. V. Kokelj & Jorgenson, 2013).

Permafrost has been mapped extensively across the Arctic, but existing products largely resolve conditions other than the mode of thaw. Many characterize the substrate, outlining permafrost extent and zonation (Brown et al., 1997), ground ice content (Jorgenson et al., 2008), and probability of occurrence (Obu et al., 2019). Others document where abrupt thaw features have formed, such as ponds and lakes (Muster et al., 2017) or retrogressive thaw slumps (Nitze et al., 2025), or quantify how far degradation of specific features has advanced (Zhang et al., 2025). A growing set of susceptibility maps predicts where specific abrupt thaw processes may develop, including thermokarst lakes (Wang et al., 2023), retrogressive thaw slumps (Makopoulou et al., 2024), and active layer detachment (Makopoulou et al., 2025), and a recent hemispheric model combines lake and slump susceptibility into a single product (Yang et al., 2025). Each of these efforts estimates the likelihood that a named landform will occur at any given point, and the combined model remains a union of predictions anchored to specific processes. None of these previous efforts addresses the question of thaw mode: whether the conditions at a site resemble those under which abrupt or gradual thaw has been observed, independent of

any single landform. Across Alaska, that contrast is carried only by point-wise samples from expert analysis of field and remote sensing observations (Webb et al., 2026), and not yet mapped as a continuous, predictive surface.

Here, we map thaw mode as a continuous, predictive surface across Alaska. We train a gradient-boosted classifier on 19,288 expert-labeled thaw points from the Alaska Permafrost Thaw Database (v2.0) (Webb et al., 2026). The classifier evaluates sites along a gradient of how strongly key environmental variables resemble either abrupt or non-abrupt thaw, using 70 geospatial indicators sourced from remote sensing, community data products, and model reanalysis. To mitigate sampling bias inherent to the Thaw Database, we evaluate the model under spatial cross-validation, bounded by an applicability mask that indicates where points fall within the envelope of the training data, and then apply it to a statewide feature grid populated by the same data sources. The resulting product is a prior-free, log-evidence index of thaw mode. We then use grouped Shapley (SHAP) values (Lundberg & Lee, 2017; Lundberg et al., 2020) to identify which geospatial indicators best distinguish between thaw modes and how they shift the classifier’s prediction. This paper describes the expert-labeled points and feature stack in Section 2, model construction and evaluation in Section 3, the susceptibility map and its drivers in Section 4, and limitations, interpretations, and comparisons with the broader permafrost mapping literature in Section 5.

2 Data

3 Methods

4 Results

5 Discussion

6 Conclusion

Open Research Statement

Statement text.

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Appendix A Appendix 1

Appendix text.